Unit 3 · Session 2 · Thu, Nov 5, 2026

GI & Nutrition Pharmacology — Session 2

4 parts · every fact cited to its slide or textbook page. About 150 minutes of focused study.

Study plan

15 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

75 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

15 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

40 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Laxatives fall into four buckets — **bulk-forming** (slowest, safest), **osmotic** (fast; PEG first-line), **stimulant** (short-term), **secretagogue** (IBS-C) — and osmotic/stimulant act in hours, bulk-forming in days. S4; S6; SG U3-R9

2. **Loperamide** and **diphenoxylate/atropine** slow the gut via mu receptors; never use them in **C. difficile** or bloody/febrile diarrhea, and atropine is there to deter misuse. S9; S11; SG U3-R10

3. IBD therapy **steps up with severity**: 5-ASA → short-term steroids → thiopurines for maintenance (test TPMT/NUDT15; cut dose with allopurinol) → biologics/small molecules. S15; S18; S19; SG U3-R11

4. Biologics share **immunosuppression risk**: screen TB and hepatitis B/C and update vaccines first; **vedolizumab** is the gut-selective one. S20; S24; SG U3-R11

5. **Fat-soluble** vitamins (A, D, E, K) accumulate; water-soluble are excreted. Long-term **PPI or metformin** → check **B12**; refeed slowly with thiamine. S30; S33; S39; SG U3-R12

6. Warfarin's effect falls with **dietary vitamin K** (pharmacodynamic) and with **St. John's wort** (CYP3A4 induction, pharmacokinetic); ginkgo and fish oil raise bleeding — always ask about supplements. S34; S35; S40; SG U3-R13

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO3.1 Identify major pharmacologic classes used in GI and nutrition conditions and describe their mechanisms of action.

Part 1 · Laxatives & Antidiarrheals
Part 2 · Inflammatory Bowel Disease
Part 3 · Nutrition, Vitamins & Dietary Supplements

IO3.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions of GI and nutrition medications.

Part 1 · Laxatives & Antidiarrheals
Part 2 · Inflammatory Bowel Disease
Part 3 · Nutrition, Vitamins & Dietary Supplements

IO3.3 Compare agents within and across these classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 1 · Laxatives & Antidiarrheals
Part 2 · Inflammatory Bowel Disease
Part 4 · Putting It All Together: GI & Nutrition Toolkit

IO3.4 Select appropriate pharmacologic therapy for common GI and nutrition conditions based on clinical presentation and patient-specific factors.

Part 1 · Laxatives & Antidiarrheals
Part 2 · Inflammatory Bowel Disease
Part 3 · Nutrition, Vitamins & Dietary Supplements
Part 4 · Putting It All Together: GI & Nutrition Toolkit

IO3.5 Identify appropriate dosing, administration, and monitoring considerations for GI and nutrition medications.

Part 2 · Inflammatory Bowel Disease
Part 3 · Nutrition, Vitamins & Dietary Supplements

IO3.6 Explain important medication-use and safety information to patients receiving GI and nutrition pharmacotherapy.

Part 1 · Laxatives & Antidiarrheals
Part 3 · Nutrition, Vitamins & Dietary Supplements

SO1 Classify laxatives and antidiarrheal agents by mechanism and clinical use, including representative generic agents.

Part 1 · Laxatives & Antidiarrheals

SO2 Explain the step-wise pharmacologic management of inflammatory bowel disease, including representative generic agents at each step.

Part 2 · Inflammatory Bowel Disease

SO3 Identify major pharmacologic classes used in nutrition support and describe clinically significant drug-nutrient and herbal-supplement interactions.

Part 3 · Nutrition, Vitamins & Dietary Supplements

SO4 Identify key contraindications, adverse effects, and drug interactions throughout.

Part 1 · Laxatives & Antidiarrheals
Part 2 · Inflammatory Bowel Disease
Part 3 · Nutrition, Vitamins & Dietary Supplements

SO5 Apply the full GI & Nutrition pharmacology toolkit to integrated clinical scenarios.

Part 4 · Putting It All Together: GI & Nutrition Toolkit

ABOUT THIS LECTURE

- PA Pharmacology Course · Lecture 3 · Session 2 of 2 — GI & Nutrition Pharmacology. Session 2: Bowel Disorders, IBD, and Nutrition/Dietary Supplements (Thursday, November 5, 8:30–11:00 AM). S1
- Session 1 followed acid and vomiting. Today follows three new pathways: how the bowel moves water, how we step up IBD therapy, and how what patients take on their own changes their medicines. S1 (notes)
- Recap: Session 1 (Nov 3) covered acid-peptic disease pharmacotherapy and antiemetics. Today picks up with lower-GI motility (Section 4) and runs through IBD, nutrition/dietary supplements, integration, and the cumulative recall quiz. S2; S2 (notes)
- Abbreviations used in the drug profiles: DR delayed-release; ER/XR extended-release; OTC over the counter; PO by mouth; PR by rectum; IV intravenous; IM intramuscular; SC subcutaneous; BID/TID/QID; CrCl creatinine clearance; CKD chronic kidney disease; CNS central nervous system; IBD inflammatory bowel disease; UC ulcerative colitis; IBS-C irritable bowel syndrome with constipation; CIC chronic idiopathic constipation; TNF tumor necrosis factor; JAK Janus kinase; TPMT/NUDT15 thiopurine-metabolizing enzymes (tested before thiopurines); G6PD; INR international normalized ratio; PCC prothrombin complex concentrate; MACE major adverse cardiovascular events; PML progressive multifocal leukoencephalopathy; CYP/P-gp; SSRI; NTD neural tube defect; EPA/DHA omega-3 fatty acids; C-V DEA Schedule V. The appendix gives formulation, dose, indications, adverse effects, contraindications and toxicity for each drug in this lecture. S53; S53 (notes)
- **Original figures in this deck:** Figures 2, 4, 7, 8 and 10 (Laxative Onset of Action by Mechanism; Step Therapy in Inflammatory Bowel Disease; Biologic & Targeted Therapies in IBD: Where Each Class Acts; Fat-Soluble vs. Water-Soluble Vitamins: Storage and Toxicity Risk; Vitamin K and CYP3A4 Inducers: A Classic Drug-Nutrient Interaction) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts. They are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. **Mechanism illustrations** (AI-generated, illustrative, not reproduced from copyrighted sources; schematic and not to scale): laxative classes (Figure 1, slide 5); loperamide and diphenoxylate (Figure 3, slide 10); 5-ASA (Figure 5, slide 17); azathioprine and 6-MP (Figure 6, slide 19); vitamin B12 absorption (Figure 9, slide 33); warfarin and vitamin K cycle (Figure 11, slide 35). Figures 1, 3, 5, 6, 9 and 11 are AI-generated and schematic. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S85; S85 (notes)

SLIDE DISCREPANCIES

- **Diphenoxylate/atropine age limit:** the speaker notes say under 2; the depth slide and drug profile say under 6. Use under 6 (the profile). S9 (notes); S10; S63
- **Thiopurine onset:** slides say weeks to months (8–12 weeks on the profile); the textbook gives a median of 17 weeks. Either way, not for acute flares. S18; S68; p. 1175

1 PART 1 · DRUG CLASS

Laxatives & Antidiarrheals

SO1 SO4 IO3.1 IO3.2 IO3.3 IO3.4 IO3.6

CLASS

- **Opposite problems, each with its own family of mechanisms.** Four laxative buckets: bulk-forming, osmotic, stimulant, secretagogue — sort any laxative name into one of them. S3; S3 (notes); S4 (notes); S5 (notes)
- **Bulk-forming** (psyllium, methylcellulose): absorb water, increase stool bulk and transit; slowest onset; generally safest for long-term use. S4
- **Osmotic** (polyethylene glycol, lactulose, magnesium hydroxide): draw water into the lumen; fast, reliable onset. S4
- **Stimulant** (senna, bisacodyl): directly stimulate enteric nerves/motility; reserve for short-term or breakthrough use. S4
- **Secretagogues** (lubiprostone, linaclotide): increase intestinal chloride and fluid secretion; used for chronic constipation, IBS-C. S4; S5

PROTOTYPE + OTHER DRUGS NAMED

- Polyethylene glycol 3350 — osmotic; first-line for chronic constipation. S5
- Psyllium — bulk-forming. Senna — stimulant. Linaclotide — secretagogue. S4
- Loperamide — antidiarrheal (peripheral opioid agonist). S9

MECHANISM OF ACTION

- Laxative mechanisms: bulk-forming holds water and expands stool; osmotic draws water into the lumen; stimulant activates enteric nerves; secretagogue increases chloride and fluid secretion → softer stool, faster transit. S5
- **Loperamide — a peripheral opioid-receptor agonist that slows GI motility without significant CNS (opioid) effects at normal doses.** S9

INDICATIONS

- **PEG is first-line for chronic constipation**; magnesium laxatives need caution in CKD. S5
- Onset (Figure 2, illustrative): osmotic ~2 h and stimulant ~8 h act fastest; secretagogue ~1 day; stool softener ~2 days and bulk-forming ~60 h take longest. **A patient who needs relief tonight is not a bulk-forming candidate.** S6; S6 (notes); SG U3-R9

- Opioid brake on the gut: opioid agonist binds mu receptors on enteric neurons → less acetylcholine release → slower peristalsis, longer transit → more water absorbed, firmer stool. S10; S10 (notes)
- Diphenoxylate/atropine — a combination opioid-receptor agonist; **atropine is included at a subtherapeutic dose specifically to discourage misuse/overdose**. S9; SG U3-R10
- Bismuth subsalicylate — antisecretory, mild antimicrobial, and anti-inflammatory effects; also used for dyspepsia. S9
- TB Opioid receptor antagonists (methylnaltrexone, naloxegol, naldemedine, alvimopan) block gut μ-receptors without entering the CNS, so analgesia is kept. p. 1166

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Stimulants cause cramping and are for short-term use. Lubiprostone causes nausea; linaclotide causes diarrhea. S4 (notes); S5
- Loperamide enters the CNS poorly (P-gp efflux), but **very high doses can cause cardiac toxicity (QT prolongation)**. S9 (notes); S10
- Bismuth subsalicylate: salicylate warning; can darken stool and tongue. S9 (notes); S10
- TB Mineral oil: aspiration causes lipid pneumonitis; long-term use impairs fat-soluble vitamin absorption. p. 1164
- TB Sodium phosphate purgatives risk electrolyte shifts and acute renal failure — avoid in frail, elderly or renal patients. p. 1165

CONTRAINDICATIONS

- **Avoid antimotility agents (loperamide, diphenoxylate/atropine) in suspected infectious/inflammatory diarrhea (e.g., C. difficile, invasive bacterial infection) — can worsen outcomes by delaying toxin/pathogen clearance.** Also avoid in bloody or febrile diarrhea. S9; S9 (notes); S10; SG U3-R10
- Bulk-forming agents need adequate fluids and are avoided in obstruction. Linaclotide: contraindicated in mechanical obstruction and in children under 2; secretagogues avoided in mechanical obstruction. S4 (notes); S5
- Diphenoxylate/atropine is **Schedule V**; avoided in young children (the slides give under 2 and under 6; the profile says under 6). S9 (notes); S10; S63
- TB Antidiarrheals should not be used with bloody diarrhea, high fever, or systemic toxicity. p. 1166

DRUG INTERACTIONS

- Psyllium: separate from other drugs by 2 hours. S55
- Bisacodyl: do not crush; avoid with antacids or milk. S59

PHARMACOKINETICS

- Loperamide has limited CNS entry because of P-gp efflux. S9 (notes)
- TB Senna works in 6–12 hours orally; bisacodyl in 6–10 hours orally and 30–60 minutes rectally. p. 1165

MONITORING

- Slowing transit in an infectious or inflammatory process can worsen the picture (toxic megacolon risk in C. difficile) — “stopping diarrhea” is not always the right first move. S9 (notes); S11 (notes)

CLINICAL PEARL / WHY

- Case: profuse watery diarrhea and fever three days after clindamycin. **Is loperamide an appropriate choice for symptomatic relief in this patient?** No — think C. difficile; avoid antimotility agents and treat the infection (oral vancomycin or fidaxomicin). S11; S11 (notes)

Step by step

Diarrhea: is an antimotility agent safe? S9; S11

Bloody stool, fever, or recent antibiotics? →

Yes → suspect C. difficile / invasive infection →

Avoid loperamide and diphenoxylate →

Treat the cause (oral vancomycin or fidaxomicin) →

No red flags → loperamide is reasonable

Figure 1 · Laxatives: how each class works. Schematic (illustrative): colon lumen cross-section — bulk-forming (psyllium) holds water and expands stool; osmotic (PEG, lactulose, magnesium) draws water into the lumen; stimulant (senna, bisacodyl) activates enteric nerves; secretagogue (lubiprostone, linaclotide) increases chloride and fluid secretion → softer stool, faster transit. S5; S5 (notes)

Figure 2 · Laxative onset of action by mechanism. Original illustrative comparison (illustrative onset-of-action values, not from Katzung; not trial data): bulk-forming 60 hr, stool softener 2 d, osmotic 2 hr, stimulant 8 hr, secretagogue (e.g., linaclotide) 1 d — osmotic and stimulant laxatives act fastest, while bulk-forming agents and stool softeners take the longest to produce an effect. S6; S6 (notes)

Figure 3 · Loperamide and diphenoxylate: opioid brake on the gut. Schematic (illustrative): loperamide binds mu-opioid receptors on myenteric neurons, reducing acetylcholine release and slowing peristalsis. S10; S10 (notes)

Laxative classes

Class	Examples	How	Use / caution
Bulk-forming	Psyllium, methylcellulose	Hold water, expand stool	Slowest, safest long term; fluids; no obstruction
Osmotic	PEG, lactulose, magnesium hydroxide	Draw water into lumen	Fast; PEG first-line chronic; magnesium caution in CKD
Stimulant	Senna, bisacodyl	Activate enteric nerves	Short-term/breakthrough; cramping
Secretagogue	Lubiprostone, linaclotide	↑ Chloride and fluid secretion	Chronic constipation, IBS-C; no obstruction

How to tell them apart. Sources: S4; S5; SG U3-R9

Laxative drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Polyethylene glycol 3350 (MiraLAX)	**Osmotic laxative**	Powder for solution (17 g scoop); large-volume bowel-prep kits	17 g in 4-8 oz fluid PO once daily; OTC use up to 7 days	Chronic and occasional constipation; bowel preparation (higher volumes)

Drug	Class	Formulation	Adult dose	Indications
Psyllium (Metamucil)	**Bulk-forming laxative**	Powder, capsules, wafers	About 3.4-6 g PO 1-3 times daily with at least 8 oz water	Constipation, IBS, mild cholesterol lowering
Methylcellulose (Citrucel)	**Bulk-forming laxative**	Powder; caplets 500 mg	About 2 g PO in 8 oz water 1-3 times daily; onset 12-72 hours	Constipation, stool bulking
Lactulose	**Osmotic laxative, ammonia-lowering agent**	Oral solution 10 g/15 mL	Constipation: 15-30 mL PO daily; hepatic encephalopathy: 30-45 mL PO 3-4 times daily, titrated to 2-3 soft stools	Constipation, hepatic encephalopathy
Senna (Senokot)	**Stimulant laxative**	Tablet 8.6 mg sennosides; liquid	8.6-17.2 mg PO at bedtime (max 34 mg/day); onset 6-12 hours	Short-term constipation, opioid-induced constipation
Bisacodyl (Dulcolax)	**Stimulant laxative**	DR tablet 5 mg; suppository 10 mg	5-15 mg PO at bedtime; 10 mg PR once daily; PO onset 6-12 hours, PR 15-60 min	Constipation, bowel preparation
Lubiprostone (Amitiza)	**Chloride channel (ClC-2) activator**	Capsule 8/24 mcg	Chronic constipation: 24 mcg PO BID with food; IBS-C (women): 8 mcg BID	Chronic idiopathic constipation, IBS-C, opioid-induced constipation
Linaclotide (Linzess)	**Guanylate cyclase-C agonist**	Capsule 72/145/290 mcg	CIC: 72-145 mcg PO daily; IBS-C: 290 mcg daily; 30 minutes before the first meal	Chronic idiopathic constipation, IBS-C

How to tell them apart. Sources: S54; S55; S56; S57; S58; S59; S60; S61

Laxative drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Polyethylene glycol 3350 (MiraLAX)	Bloating, nausea, gas, loose stools	Bowel obstruction, perforation, ileus, toxic megacolon	Dehydration and electrolyte loss with excess or bowel-prep volumes; rare allergic reactions
Psyllium (Metamucil)	Bloating, gas, abdominal fullness	Bowel obstruction, fecal impaction, dysphagia	Esophageal or intestinal obstruction if taken with too little fluid; allergy; separate from other drugs by 2 hours
Methylcellulose (Citrucel)	Bloating, gas	Bowel obstruction, dysphagia	Obstruction if taken without adequate fluid; may slow absorption of other drugs

Drug	Adverse effects	Contraindications	Toxicity / overdose
Lactulose	Gas, bloating, cramps, diarrhea	Galactosemia, bowel obstruction	Dehydration and hypernatremia with over-use; worsens diabetes control slightly
Senna (Senokot)	Cramping, diarrhea, urine discoloration, melanosis coli	Bowel obstruction, appendicitis, undiagnosed abdominal pain	Hypokalemia and dehydration with chronic use; laxative dependence
Bisacodyl (Dulcolax)	Cramping, nausea, rectal irritation	Bowel obstruction, appendicitis, acute abdomen	Do not crush; avoid with antacids or milk; excess causes diarrhea, hypokalemia
Lubiprostone (Amitiza)	Nausea, diarrhea, headache, abdominal pain	Mechanical GI obstruction	Dizziness, syncope or dyspnea (rare); overdose: nausea, diarrhea; avoid in pregnancy unless needed
Linaclotide (Linzess)	Diarrhea, abdominal pain, gas, bloating	Boxed: children under 2 years; mechanical GI obstruction	Severe dehydration (deaths in young children); severe diarrhea; stop if severe or prolonged

How to tell them apart. Sources: S54; S55; S56; S57; S58; S59; S60; S61

Antidiarrheal drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Loperamide (Imodium)	**Peripheral mu-opioid agonist**	Capsule 2 mg; tablet 2 mg; oral liquid	4 mg PO first dose, then 2 mg after each loose stool; max 16 mg/day Rx, 8 mg/day OTC	Acute and chronic diarrhea, traveler's diarrhea
Diphenoxylate/atropine (Lomotil)	**Opioid antidiarrheal (C-V) plus anticholinergic**	Tablet 2.5 mg/0.025 mg; oral liquid 2.5 mg/5 mL	5 mg (2 tablets) PO QID initially; max 20 mg/day; reduce as controlled	Adjunct for diarrhea

How to tell them apart. Sources: S62; S63

Antidiarrheal drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Loperamide (Imodium)	Constipation, cramps, dizziness, nausea	Bloody stool or high fever, C. difficile, children under 2, ileus	High doses: QT prolongation, torsades, cardiac arrest; CNS depression in overdose (naloxone, repeated)

Drug	Adverse effects	Contraindications	Toxicity / overdose
Diphenoxylate/atropine (Lomotil)	Drowsiness, dry mouth, tachycardia, urinary retention	Children under 6 years, obstructive jaundice, C. difficile or infectious diarrhea	Respiratory depression (especially children) and anticholinergic toxicity; naloxone, with monitoring for relapse

How to tell them apart. Sources: S62; S63

Other bowel drugs in the reading (textbook)

Group	Drugs	Point
Bulk-forming laxatives	Psyllium, methylcellulose, polycarbophil	Hydrophilic gel distends colon; bloating, flatus
Stool surfactant agents (softeners)	Docusate, glycerin, mineral oil	Mineral oil: lipid pneumonitis if aspirated
Osmotic laxatives; balanced polyethylene glycol	Magnesium, sorbitol, lactulose; PEG lavage	PEG causes no significant cramps or flatus; bowel prep 1–2 L
Stimulant laxatives: anthraquinone and diphenylmethane derivatives	Senna, cascara (melanosis coli); bisacodyl	Long-term use probably safe in most
Chloride secretion activators	Lubiprostone (ClC-2); linaclotide, plecanatide (GC-C)	Nausea up to 30% (lubiprostone); diarrhea up to 20% (linaclotide)
Opioid receptor antagonists	Methylnaltrexone, naloxegol, naldemedine, alvimopan	Opioid-induced constipation; alvimopan postoperative ileus only
Serotonin 5-HT4-receptor agonists	Prucalopride	Chronic constipation; 1 mg/day if CrCl < 30
Antidiarrheal agents: opioid agonists	Loperamide, diphenoxylate, eluxadoline	Eluxadoline: IBS-D; sphincter of Oddi spasm
Colloidal bismuth compounds	Bismuth subsalicylate	Traveler’s diarrhea
Bile salt-binding resins	Cholestyramine, colestipol, colesevelam	Bile-acid diarrhea; separate other drugs by 2 hours
Octreotide	Somatostatin analog	Secretory tumor diarrhea; gallstones with long-term use
IBS: antispasmodics (anticholinergics)	Dicyclomine, hyoscyamine	Short-term pain relief; anticholinergic effects
IBS: serotonin 5-HT3-receptor antagonists	Alosetron	Women with severe IBS-D; ischemic colitis, severe constipation
IBS: chloride channel activators	Lubiprostone 8 mcg twice daily; linaclotide 290 mcg	IBS-C

How to tell them apart. Sources: p. 1164; p. 1165; p. 1166; p. 1167; p. 1168

SLIDE DISCREPANCY

- **Age cutoff for diphenoxylate/atropine**
- Slides: Avoid in children under 2 (S9 notes); under 6 (S10, S63). S9 (notes); S10; S63
- Textbook: Small amounts of atropine are added to discourage overdosage (no age cutoff given). p. 1166

WHY IT HAPPENS

- Why is atropine in Lomotil? At the tiny dose given it does nothing — but anyone taking big doses to get high gets unpleasant anticholinergic effects. S9; SG U3-R10
- Why not loperamide in C. difficile? Slowing transit keeps toxin in the colon longer — toxic megacolon risk. S9 (notes); SG U3-R10

HIGH-YIELD

- Four laxative buckets: bulk-forming (slowest, safest), osmotic (fast; PEG first-line), stimulant (short-term), secretagogue (IBS-C). S4; S5
- No antimotility agents in C. difficile or bloody/febrile diarrhea. S9; SG U3-R10

2 PART 2 · DRUG CLASS

Inflammatory Bowel Disease

SO2 SO4 IO3.1 IO3.2 IO3.3 IO3.4 IO3.5

CLASS

- **Step therapy, escalating with disease severity:** 5-ASA agents for mild disease → short-term corticosteroids for flares → immunomodulators for maintenance → biologics/small molecules for moderate-to-severe disease. S14; S14 (notes); S15; SG U3-R11
- This escalation ladder is the organizing framework for every IBD drug — ask where each drug "lives" on it. S15 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Mesalamine — 5-ASA (step 1). S16
- Prednisone, budesonide — corticosteroids (flares). S18
- Azathioprine, 6-mercaptopurine — immunomodulators (maintenance). S18
- Infliximab, adalimumab, vedolizumab, ustekinumab, tofacitinib — targeted therapy. S20

MECHANISM OF ACTION

- **Mesalamine, sulfasalazine — anti-inflammatory action local to the GI mucosa.** Sulfasalazine reaches the colon → colonic bacteria cleave the azo bond → 5-ASA (mesalamine) plus sulfapyridine released → 5-ASA acts on mucosa, reducing inflammatory mediators → less mucosal inflammation. S16; S17; S17 (notes)
- Azathioprine is converted to 6-mercaptopurine → metabolized to thioguanine nucleotides → block purine synthesis in lymphocytes → fewer proliferating T and B cells → slow, sustained remission. S19; S19 (notes)

INDICATIONS

- 5-ASA: first-line for mild-to-moderate ulcerative colitis; more limited role in Crohn disease. S16
- Corticosteroids: short-term induction of remission during an acute flare; not appropriate for long-term maintenance given cumulative toxicity. S18
- Immunomodulators: long-term maintenance of remission; onset is slow (weeks to months) — not useful for acute flares. S18
- Delayed-release forms target the ileum or colon; enemas and suppositories treat distal disease. S16 (notes); S17
- TB Corticosteroids: prednisone 40–60 mg/d for flares, tapered after response; rectal forms for distal disease; not for maintenance. p. 1174; p. 1175

- **Targeted therapy:** TNF-α inhibitors block tumor necrosis factor-alpha (infliximab, adalimumab); the integrin inhibitor blocks α4β7 integrin-mediated leukocyte gut-homing (vedolizumab); the IL-12/23 inhibitor blocks interleukin-12 and -23 signaling (ustekinumab); the JAK inhibitor blocks intracellular JAK signaling (tofacitinib). S20; S20 (notes); S21
- Vedolizumab is the most **gut-selective** agent — less systemic immunosuppression than TNF inhibitors, JAK inhibitors, or thiopurines. S52; SG U3-R11
- TB **Chemistry & formulations:** azo compounds (sulfasalazine, balsalazide, olsalazine) are cleaved by colonic bacteria; mesalamine formulations release 5-ASA at set pH or times. p. 1173

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **The major safety theme across immunomodulators and biologics: immunosuppression → increased infection risk.** S24; S24 (notes)
- Sulfasalazine: the sulfapyridine component causes most adverse effects — rash, hemolysis in G6PD deficiency, agranulocytosis, folate deficiency (give folic acid); caution in sulfa allergy. S16; S16 (notes); S17
- Mesalamine: monitor renal function (interstitial nephritis). 5-ASA agents are generally well tolerated — a favorable long-term safety profile relative to steroids and immunomodulators. S16; S16 (notes); S17
- Thiopurines: increased infection and lymphoma risk; monitor CBC and liver tests (bone marrow suppression, hepatotoxicity). S18; S19
- TB Up to 40% cannot tolerate therapeutic sulfasalazine doses; give 1 mg/d folic acid; reversible oligospermia. p. 1174

CONTRAINDICATIONS

- Sulfasalazine: caution in sulfa allergy (sulfapyridine). S16
- TB No azathioprine/6-MP with absent TPMT activity (low in 11%, absent in 0.3%). p. 1175

- **TB** Anti-TNF: TB reactivation — test before starting; antibodies to the drug reduce response; infliximab infusion reactions in up to 10%. p. 1177
- **TB** Tofacitinib: fourfold herpes zoster reactivation (vaccinate first); check lipids at 8 weeks. p. 1178

DRUG INTERACTIONS

- **Allopurinol blocks thiopurine metabolism — reduce the thiopurine dose by about two-thirds to three-quarters**; febuxostat is avoided. S19; S84
- Ketoconazole (CYP3A4 inhibitor) raises budesonide exposure. S66; S84

PHARMACOKINETICS

- Budesonide has high first-pass metabolism — less systemic exposure than prednisone. S18; S18 (notes)
- **TB** Thiopurines take a median 17 weeks to benefit; budesonide bioavailability about 10% (CYP3A4 inhibitors raise it). p. 1174; p. 1175

MONITORING

- **Test TPMT and NUDT15 before starting thiopurines; low activity raises myelosuppression risk.** S19; S19 (notes)
- **Before biologics:** tuberculosis and hepatitis B/C screening; update vaccines, especially live vaccines, before starting; regular labs to catch cytopenias, hepatotoxicity, or infection early. S24; S24 (notes)

CLINICAL PEARL / WHY

- Infliximab and adalimumab are the same TNF-α inhibitors as the RA biologic DMARDs in Lecture 2 — same mechanism, same risk profile in a new organ system. S20 (notes); S24; S24 (notes)
- "Fast but not for the long term" (steroids) vs. "slow but sustainable" (immunomodulators) parallels the acute-vs-chronic gout framework. S18 (notes)
- Case: newly diagnosed mild UC asks why mesalamine, not a biologic. **How would you explain the step-therapy rationale to this patient?** S25
- Therapy is matched to severity: mild disease responds to 5-ASA with a favorable safety profile; biologics are reserved for moderate-to-severe or refractory disease given cost, administration burden, and immunosuppression risks. S25; S25 (notes)
- Name stems: -ximab chimeric (infliximab, anti-TNF-α); -umab human (adalimumab, ustekinumab); -zumab humanized (vedolizumab, α4β7 integrin). S28
- More stems: -tinib kinase inhibitor (tofacitinib, JAK); -sone corticosteroid (prednisone, short-term induction); -tide peptide (linaclotide, GC-C agonist). 5-ASA agents, thiopurines and opioid antidiarrheals share no stem. A stem is a memory aid, not a rule. S28; S28 (notes)

Step by step

IBD step therapy S15; S18; S19; S24

Mild UC → mesalamine → Flare → short-term prednisone or budesonide →

Maintenance → azathioprine/6-MP (TPMT/NUDT15 first) →

Moderate-severe or refractory → biologic or JAK inhibitor →

Before biologics: TB, hepatitis B/C, vaccines

Figure 4 · Step therapy in inflammatory bowel disease. Original illustrative diagram (not from Katzung): therapy escalates from 5-ASA agents for mild disease, through short-term corticosteroids for flares and immunomodulators for maintenance, to biologics/small molecules for moderate-to-severe disease. S15; S15 (notes)

Figure 5 · 5-ASA: local anti-inflammatory action. Schematic (illustrative): sulfasalazine is cleaved by colonic bacteria; the released 5-ASA acts locally on the inflamed colonic mucosa. S17; S17 (notes)

Figure 6 · Azathioprine and 6-MP: purine antimetabolites. Schematic (illustrative): thiopurine metabolites block lymphocyte purine synthesis, so fewer lymphocytes proliferate. S19; S19 (notes)

Figure 7 · Biologic & targeted therapies in IBD: where each class acts. Original illustrative diagram (not from Katzung): infliximab/adalimumab block TNF-α, vedolizumab blocks leukocyte gut-homing, ustekinumab blocks IL-12/23, and tofacitinib blocks JAK signaling — four distinct points of intervention in chronic mucosal inflammation. S21; S21 (notes)

IBD drug classes

Step	Drugs	Role	Key safety
1 · 5-ASA	Mesalamine, sulfasalazine	Mild-moderate UC (induction + maintenance)	Sulfapyridine AEs; renal monitoring (mesalamine)
Flare	Prednisone, budesonide	Short-term induction	Cumulative toxicity; not maintenance
Maintenance	Azathioprine, 6-MP	Slow, sustained remission	TPMT/NUDT15; CBC, LFTs; allopurinol interaction
Targeted	Infliximab, adalimumab, vedolizumab, ustekinumab, tofacitinib	Moderate-severe or refractory	Infection: TB/hepatitis screen, vaccines first

How to tell them apart. Sources: S15; S16; S18; S19; S20; S24

Name clues: bowel disorders and IBD

Stem	Drug class	Examples	Key point
-ximab	Monoclonal Ab (chimeric)	Infliximab	Anti-TNF-α
-umab	Monoclonal Ab (human)	Adalimumab, ustekinumab	Anti-TNF-α; anti-IL-12/23

Stem	Drug class	Examples	Key point
-zumab	Monoclonal Ab (humanized)	Vedolizumab	Blocks α4β7 integrin
-tinib	Kinase inhibitor	Tofacitinib	JAK inhibitor
-sone	Corticosteroid	Prednisone	Short-term induction
-tide	Peptide	Linaclotide	GC-C agonist
None shared	5-ASA agents	Mesalamine, sulfasalazine	Local mucosal action
None shared	Thiopurines	Azathioprine, mercaptopurine	Slow maintenance
None shared	Opioid antidiarrheals	Loperamide, diphenoxylate	Slows motility

How to tell them apart. Sources: S28; S28 (notes)

5-ASA, steroid and thiopurine profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Mesalamine (Lialda, Asacol HD, Pentasa)	**5-ASA aminosalicylate**	DR tablet 800 mg/1.2 g; capsule; suppository 1,000 mg; enema 4 g/60 mL	Mild-moderate UC: 2.4-4.8 g/day PO once daily or divided; 1,000 mg PR at bedtime for proctitis	Induction and maintenance in ulcerative colitis; mild Crohn disease (some forms)
Sulfasalazine (Azulfidine)	**Sulfapyridine plus 5-ASA prodrug**	Tablet 500 mg; EN-tabs DR 500 mg	Active UC: start 1-2 g/day, titrate to 3-4 g/day in divided doses; maintenance 2 g/day; folic acid 1 mg daily	Ulcerative colitis, rheumatoid arthritis (EN-tabs)
Budesonide (Entocort EC, Uceris)	**Locally acting corticosteroid**	Capsule 3 mg; ER tablet 9 mg; foam	Crohn disease: 9 mg PO every morning for up to 8 weeks; UC: 9 mg ER tablet daily for 8 weeks	Mild-moderate ileocecal Crohn disease, mild-moderate UC
Prednisone	**Systemic corticosteroid**	Tablet 1/2.5/5/10/20/50 mg; oral solution	Flare: 40-60 mg PO daily, then taper over weeks; short-term induction only	Induction of remission in IBD flares; many inflammatory conditions
Azathioprine (Imuran)	**Purine antimetabolite**	Tablet 50/75/100 mg; IV	IBD maintenance: 1.5-2.5 mg/kg PO daily after TPMT/NUDT15 testing; onset 8-12 weeks	Maintenance of remission in Crohn disease and UC (off-label), organ transplant

Drug	Class	Formulation	Adult dose	Indications
Mercaptopurine (6-MP, Purinethol)	**Purine antimetabolite**	Tablet 50 mg; oral suspension	IBD (off-label): 1-1.5 mg/kg PO daily after TPMT/NUDT15 testing	Maintenance in Crohn disease and UC; acute lymphoblastic leukemia

How to tell them apart. Sources: S64; S65; S66; S67; S68; S69

5-ASA, steroid and thiopurine profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Mesalamine (Lialda, Asacol HD, Pentasa)	Headache, nausea, abdominal pain, flatulence	Salicylate hypersensitivity	Interstitial nephritis (monitor creatinine); acute intolerance syndrome (cramps, bloody diarrhea); hepatotoxicity rare
Sulfasalazine (Azulfidine)	Nausea, headache, rash, orange urine, reversible low sperm count	Sulfa or salicylate allergy, intestinal or urinary obstruction, porphyria	Agranulocytosis, hemolysis in G6PD deficiency, Stevens-Johnson syndrome, hepatitis; folate deficiency
Budesonide (Entocort EC, Uceris)	Headache, nausea, acne, adrenal suppression with long courses	Hypersensitivity	Hypercortisolism with CYP3A4 inhibitors (ketoconazole, grapefruit juice); infection risk
Prednisone	Hyperglycemia, insomnia, mood change, weight gain, infection risk	Systemic fungal infection; live vaccines at high dose	Adrenal suppression and adrenal crisis on abrupt stopping; osteoporosis, cataracts, Cushingoid features
Azathioprine (Imuran)	Nausea, leukopenia, hepatotoxicity, pancreatitis, infection	Hypersensitivity; prior alkylating therapy; use with febuxostat	Myelosuppression (higher with low TPMT), lymphoma and hepatosplenic T-cell lymphoma; allopurinol requires major dose cut
Mercaptopurine (6-MP, Purinethol)	Nausea, leukopenia, hepatotoxicity, pancreatitis	Hypersensitivity; use with febuxostat	Myelosuppression, hepatotoxicity, lymphoma risk; reduce dose with allopurinol

How to tell them apart. Sources: S64; S65; S66; S67; S68; S69

Biologic and JAK inhibitor profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Infliximab (Remicade)	**Chimeric anti-TNF-alpha antibody**	IV 100 mg vial; SC 120 mg syringe (Zymfentra)	Induction 5 mg/kg IV at weeks 0, 2, 6, then every 8 weeks; dose may be escalated	Crohn disease, ulcerative colitis, rheumatoid arthritis, psoriasis

Drug	Class	Formulation	Adult dose	Indications
Adalimumab (Humira)	**Human anti-TNF-alpha antibody**	SC pen or syringe 40 mg/0.4 mL	Crohn disease and UC: 160 mg SC day 1, 80 mg day 15, then 40 mg every 2 weeks	Crohn disease, ulcerative colitis, rheumatoid arthritis, psoriasis
Vedolizumab (Entyvio)	**Humanized alpha4beta7 integrin antibody**	IV 300 mg vial; SC pen 108 mg	300 mg IV at weeks 0, 2, 6, then every 8 weeks; SC 108 mg every 2 weeks after IV induction	Moderate-severe ulcerative colitis and Crohn disease
Ustekinumab (Stelara)	**Human anti-IL-12/23 antibody**	IV 130 mg/26 mL vial; SC 45/90 mg	IV induction by weight (260 mg under 55 kg, 390 mg 55-85 kg, 520 mg over 85 kg), then 90 mg SC every 8 weeks	Crohn disease, ulcerative colitis, psoriasis
Tofacitinib (Xeljanz)	**JAK inhibitor**	Tablet 5/10 mg; XR tablet 11/22 mg	UC: 10 mg PO BID for at least 8 weeks, then 5 mg BID (lowest effective dose)	Ulcerative colitis, rheumatoid arthritis, psoriatic arthritis

How to tell them apart. Sources: S70; S71; S72; S73; S74

Biologic and JAK inhibitor profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Infliximab (Remicade)	Infusion reactions, headache, infections, rash	Hypersensitivity to murine proteins; high doses in moderate-severe heart failure	TB and hepatitis B reactivation, serious infections, demyelination, hepatosplenic T-cell lymphoma; anti-drug antibodies
Adalimumab (Humira)	Injection-site reactions, infections, headache, rash	Hypersensitivity; active serious infection	TB and hepatitis B reactivation, serious infections, malignancy, demyelination, lupus-like syndrome
Vedolizumab (Entyvio)	Nasopharyngitis, headache, arthralgia, infusion reactions	Hypersensitivity	Infections and rare liver injury; PML not reported to date but vigilance advised; gut-selective with less systemic immunosuppression
Ustekinumab (Stelara)	Nasopharyngitis, injection-site reactions, headache, fatigue	Hypersensitivity	Serious infections, TB, reversible posterior leukoencephalopathy (rare), hypersensitivity
Tofacitinib (Xeljanz)	Headache, nasopharyngitis, raised lipids, herpes zoster	Hypersensitivity; severe hepatic impairment; active serious infection	Boxed: serious infections, mortality, malignancy, MACE, thrombosis; cytopenias, GI perforation; use after TNF-inhibitor failure

How to tell them apart. Sources: S70; S71; S72; S73; S74

IBD and other GI drugs in the reading (textbook)

Drug / group	Mechanism	Point
Infliximab, adalimumab, certolizumab, golimumab	Anti-TNF antibodies	Screen TB first; certolizumab lacks an Fc portion
Anti-integrin therapy: vedolizumab (natalizumab)	α4β7 (α4) integrin block	Natalizumab risks PML; vedolizumab PML risk extremely low
Anti-IL-23 therapy: ustekinumab, risankizumab	IL-12/23 p40; IL-23 p19	Risankizumab: drug-induced liver injury (check LFTs)
Janus kinase inhibitors: tofacitinib, upadacitinib	JAK-STAT	Boxed: heart attack, stroke, thrombosis, cancer, death
Ozanimod	Sphingosine-1-phosphate receptor agonist	Bradyarrhythmia, macular edema; ECG first
Pancreatic enzyme supplements (pancrelipase)	Lipase, amylase, protease replacement	With every meal; swallow, don't chew
Glucagon-like peptide 2 analog for short-bowel syndrome (teduglutide)	Mucosal growth	Cuts parenteral support; neoplasia risk
Bile acid agents: ursodiol, obeticholic acid	Lower bile cholesterol; FXR ligand	Gallstone dissolution; primary biliary cirrhosis
Drugs used to treat variceal hemorrhage: somatostatin & octreotide	Reduce portal blood flow	Acute variceal bleeding for 3–5 days
Vasopressin & terlipressin	Splanchnic vasoconstriction	Vasopressin: ischemia, hyponatremia
Beta-receptor-blocking drugs (propranolol, nadolol)	Nonselective β block lowers portal inflow	Prevent variceal bleeding (15% vs 25%)

How to tell them apart. Sources: p. 1176; p. 1177; p. 1178; p. 1179; p. 1180

SLIDE DISCREPANCY

- **Thiopurine onset**
- Slides: Onset takes weeks to months (azathioprine 8–12 weeks). S19; S68
- Textbook: A median delay of 17 weeks before benefit. p. 1175

WHY IT HAPPENS

- Why is vedolizumab gut-selective? α4β7 integrin is the gut-homing address tag; block it and leukocytes can't enter the bowel, while immunity elsewhere is spared. S20; SG U3-R11
- Why give folic acid with sulfasalazine? Sulfapyridine causes folate deficiency. S16 (notes); S17

HIGH-YIELD

- IBD therapy follows a step-wise escalation matched to disease severity — from 5-ASA agents to biologics/small molecules. S48; S15
- Biologic agents share both their targeted mechanisms and their immunosuppression-related safety profile across organ systems (RA in Lecture 2, IBD here). S48; S24
- TPMT/NUDT15 before thiopurines; allopurinol → cut dose by 2/3–3/4. S19

3 PART 3 · DRUG CLASS

Nutrition, Vitamins & Dietary Supplements

SO3 SO4 IO3.1 IO3.2 IO3.4 IO3.5 IO3.6

CLASS

- **What patients take on their own can change how their medications work.** Ask one question about any vitamin: can it build up to toxic levels? S29; S29 (notes); S30 (notes)
- **Fat-soluble (A, D, E, K):** stored in adipose tissue and the liver; accumulate with excess intake — real toxicity risk; absorption depends on dietary fat and bile, impaired by fat-malabsorption conditions. S30; SG U3-R12
- **Water-soluble (B-complex, C):** not significantly stored — excreted renally; toxicity is rare except at very high supplemental doses; deficiency develops more quickly without ongoing intake. S30; SG U3-R12
- Figure 8 (illustrative storage/toxicity-risk index): vitamin A 85 and D 90 (fat-soluble) vs. C 10 and B-complex 8 (water-soluble). Vitamin D and A toxicity from OTC overuse are the clearest real examples. S31; S31 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Cyanocobalamin (B12), folic acid, vitamin D, phytonadione (K1), ferrous sulfate. S76; S77; S78; S79; S80
- St. John's wort, ginkgo biloba, fish oil — supplements with drug interactions. S40

MECHANISM OF ACTION

- **B12 absorption:** dietary B12 is bound to food protein → stomach acid and pepsin release it → free B12 binds intrinsic factor from parietal cells → the complex is absorbed in the terminal ileum. S33; S33 (notes)

INDICATIONS

- **Enteral nutrition** (oral or tube feeding) is preferred whenever the gut is functional — **"if the gut works, use it"**; lower infection/complication risk. S38; S38 (notes); S39

- Reduced vitamin K is recycled by VKORC1; **warfarin blocks VKORC1, so fewer active factors** — active vitamin K carboxylates clotting factors II, VII, IX, X — so fewer active factors. S35; S35 (notes)
- **Two ways to lower warfarin's effect:** dietary vitamin K opposes it pharmacodynamically (supports clotting factor synthesis); St. John's wort induces CYP enzymes and speeds clearance — a pharmacokinetic interaction. **Result: lower INR and less anticoagulation.** S34; S34 (notes); S35

- **Parenteral nutrition** is delivered intravenously, bypassing the GI tract entirely; reserved for a non-functional or inaccessible GI tract; requires close monitoring — electrolyte shifts, refeeding syndrome, catheter-related infection (the main parenteral risk). S38; S39
- Treat B12 deficiency with intramuscular or high-dose oral B12. S33; S33 (notes)

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Vitamin B12 deficiency:** megaloblastic anemia and neurologic symptoms/neuropathy that can become irreversible if treatment is delayed. S32; S32 (notes); S33
- **Folate:** deficiency causes megaloblastic anemia; critical in pregnancy to prevent neural tube defects. S32; S32 (notes)
- **Vitamin D:** deficiency impairs bone mineralization (rickets in children, osteomalacia in adults). **Vitamin K:** deficiency impairs synthesis of clotting factors II, VII, IX, and X, increasing bleeding risk. S32; S32 (notes)
- **Iron:** deficiency causes microcytic anemia; absorption is enhanced by vitamin C and impaired by concurrent antacid or PPI use. S32; S32 (notes)
- **Refeeding sequence in prolonged starvation:** carbohydrate feeding raises insulin → phosphate, potassium and magnesium shift into cells → dangerously low levels: heart, muscle and breathing problems. S39; S39 (notes)

CONTRAINDICATIONS

- St. John's wort: avoid with SSRIs, transplant drugs, HIV protease inhibitors, oral contraceptives, warfarin. S81

DRUG INTERACTIONS

- **St. John's Wort — CYP3A4 induction; reduces effectiveness of many drugs (warfarin, oral contraceptives, some antiretrovirals).** S40; S35; SG U3-R13
- **Ginkgo biloba and fish oil — can increase bleeding risk, especially combined with anticoagulants/antiplatelets.** S40; S35
- **At risk for B12 deficiency:** long-term PPI or H2 blocker use, metformin (impairs ileal uptake), bariatric surgery, older age, vegan diets. S32; S33; S33 (notes); SG U3-R12
- TB St. John's wort activates PXR, inducing CYP 3A4, 2C9, 1A2 and P-glycoprotein: lowers digoxin, birth control, cyclosporine, HIV drugs, warfarin; serotonin syndrome with antidepressants. p. 1214
- TB Garlic, ginkgo and ginseng have antiplatelet effects — avoid or monitor with anticoagulants and around surgery. p. 1208; p. 1210; p. 1211
- TB Coenzyme Q10 and melatonin may decrease warfarin's effect; glucosamine may increase the INR. p. 1216; p. 1218

PHARMACOKINETICS

- Fat-soluble vitamin absorption depends on dietary fat and bile. S30

MONITORING

- Monitor B12 in long-term PPI or metformin users. S33
- **Keep vitamin K intake consistent rather than avoiding it.** Check INR when starting or stopping any supplement; stopping St. John's wort can raise INR. S35; S35 (notes)
- **Refeeding:** start feeding slowly in malnourished patients; check phosphate, potassium and magnesium daily at the start; give thiamine before feeding begins. S39; S39 (notes)

CLINICAL PEARL / WHY

- **Supplements are not held to the same FDA premarket efficacy/safety standards as prescription drugs — potency and purity can vary by product.** S40; S40 (notes)

- Always ask patients specifically about supplement use — many don't mention it unless asked; counsel them to disclose all supplement use to every prescriber and pharmacist. S40; S40 (notes)
- Case: warfarin for atrial fibrillation, subtherapeutic INR despite adherence; started St. John's Wort for mood three weeks ago. **How does this explain the subtherapeutic INR, and what should happen next?** S41
- CYP3A4 induction speeds warfarin metabolism — asking about supplements is a required part of every medication reconciliation. S41; S41 (notes); SG U3-R13
- The B12/PPI link and the vitamin K/clotting-factor link tie back to earlier drug classes in this unit. S32 (notes)
- **TB** Under DSHEA (1994), supplements are regulated as foods: the FDA must prove a product harmful before restricting it. p. 1204

Step by step

Refeeding sequence S39

Prolonged starvation → Carbohydrate feeding raises insulin →

Phosphate, potassium, magnesium shift into cells →

Dangerously low levels: heart, muscle, breathing →

Prevent: start slowly, thiamine first, check electrolytes daily

B12 absorption S33

Dietary B12 bound to food protein → Stomach acid and pepsin release it →

Free B12 binds intrinsic factor → Absorbed in the terminal ileum →

PPIs, metformin, ileal surgery impair it

Figure 8 · Fat-soluble vs. water-soluble vitamins: storage and toxicity risk. Original illustrative comparison (illustrative storage/toxicity-risk index, not from Katzung): fat-soluble vitamins A (85) and D (90) show substantially higher tissue-storage and toxicity-risk scores than water-soluble vitamins C (10) and B-complex (8). S31; S31 (notes)

Figure 9 · Vitamin B12 absorption: where PPIs and metformin interfere. Schematic (illustrative): vitamin B12 absorption needs acid and intrinsic factor and occurs in the ileum; PPIs, metformin and ileal surgery impair it. S33; S33 (notes)

Figure 10 · Vitamin K and CYP3A4 inducers: a classic drug-nutrient interaction. Original illustrative diagram (not from Katzung): dietary vitamin K opposes warfarin's anticoagulant effect by supporting clotting factor synthesis, while CYP3A4-inducing herbal supplements (e.g., St. John's Wort) increase warfarin metabolism — both lowering anticoagulant effect through different mechanisms (pharmacodynamic antagonism vs. pharmacokinetic induction). S34; S34 (notes)

Figure 11 · Warfarin: vitamin K cycle and herbal interactions. Schematic (illustrative): vitamin K cycle in a liver cell — warfarin blocks VKORC1, and St. John's wort speeds warfarin clearance. S35; S35 (notes)

Vitamin and mineral deficiency syndromes

Nutrient	Deficiency	Drug link
Vitamin B12	Megaloblastic anemia, neuropathy (can be irreversible)	Long-term PPI/H2 blocker, metformin, bariatric surgery
Folate	Megaloblastic anemia; neural tube defects in pregnancy	Sulfasalazine, methotrexate (give folic acid)
Vitamin D	Rickets (children), osteomalacia (adults)	Bone pharmacology unit
Vitamin K	↓ Factors II, VII, IX, X → bleeding	Warfarin opposes it
Iron	Microcytic anemia	Antacids/PPIs ↓ absorption; vitamin C ↑

How to tell them apart. Sources: S32; S33; S77; SG U3-R12

Enteral vs. parenteral nutrition

	Enteral	Parenteral
Route	GI tract (oral or tube feeding)	Intravenous, bypassing the GI tract
Use when	Gut is functional (“if the gut works, use it”)	Gut non-functional or inaccessible
Main risk	Lower infection/complication risk	Catheter infection; electrolyte shifts; refeeding syndrome

How to tell them apart. Sources: S38; S39

Warfarin, vitamin and mineral profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Warfarin (Coumadin, Jantoven)	**Vitamin K antagonist (VKORC1 inhibitor)**	Tablet 1/2/2.5/3/4/5/6/7.5/10 mg	Start 2-5 mg PO daily, adjust to INR goal (usually 2-3)	Venous thromboembolism, atrial fibrillation, mechanical heart valves
Cyanocobalamin (vitamin B12)	**Vitamin**	Tablet 100-1,000 mcg; IM injection 1,000 mcg/mL; nasal gel	Deficiency: 1,000 mcg IM daily for 1 week, then weekly, then monthly; or 1,000-2,000 mcg PO daily	B12 deficiency, pernicious anemia, PPI or metformin-related deficiency
Folic acid (vitamin B9)	**Water-soluble vitamin**	Tablet 0.4/0.8/1 mg; injection	Deficiency 1 mg PO daily; 0.4 mg daily for neural tube defect prevention (4 mg if prior NTD)	Megaloblastic anemia, pregnancy, sulfasalazine and methotrexate therapy

Drug	Class	Formulation	Adult dose	Indications
Vitamin D (cholecalciferol)	**Fat-soluble vitamin**	Capsule/tablet 400-5,000 IU; 50,000 IU capsule	Maintenance 600-800 IU PO daily; deficiency 50,000 IU weekly for about 8 weeks	Rickets, osteomalacia, osteoporosis support, deficiency
Phytonadione (vitamin K1)	**Fat-soluble vitamin**	Tablet 5 mg; injection 10 mg/mL	Warfarin over-anticoagulation: 1-10 mg PO or slow IV by INR; newborn prophylaxis 0.5-1 mg IM	Warfarin reversal, vitamin K deficiency bleeding, malabsorption
Ferrous sulfate	**Iron supplement**	Tablet 325 mg (65 mg elemental); liquid	325 mg PO once daily or every other day, ideally with vitamin C	Iron-deficiency anemia

How to tell them apart. Sources: S75; S76; S77; S78; S79; S80

Warfarin, vitamin and mineral profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Warfarin (Coumadin, Jantoven)	Bleeding, bruising, hair loss, skin necrosis (rare)	Pregnancy (most cases), active bleeding, recent CNS surgery, uncontrolled hypertension	Hemorrhage; reversal with vitamin K and 4-factor PCC; many interactions (St. John's wort lowers INR)
Cyanocobalamin (vitamin B12)	Injection-site pain, mild diarrhea, itching; early low potassium when treating severe anemia	Cobalt allergy, Leber hereditary optic neuropathy	Low toxicity; excess is excreted; folate alone can mask B12 anemia while neuropathy progresses
Folic acid (vitamin B9)	Nausea, bloating, rash (rare)	Untreated B12 deficiency (can mask anemia)	Low toxicity; high doses may mask B12 deficiency and worsen neurologic injury
Vitamin D (cholecalciferol)	Usually none; nausea and constipation with excess	Hypercalcemia, hypervitaminosis D	Toxicity: hypercalcemia, kidney stones, calcification; fat-soluble so accumulates
Phytonadione (vitamin K1)	Injection-site reactions; flushing	Hypersensitivity	IV: boxed warning for anaphylaxis (infuse slowly); overcorrection causes warfarin resistance
Ferrous sulfate	Constipation, nausea, dark stools, abdominal pain	Hemochromatosis, hemosiderosis, hemolytic anemia	Acute iron overdose is a pediatric emergency: GI bleeding, shock, liver failure; chelate with deferoxamine

How to tell them apart. Sources: S75; S76; S77; S78; S79; S80

Herbal and dietary supplement profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
St. John's wort (Hypericum perforatum)	**Herbal supplement (CYP/P-gp inducer)**	Capsule/tablet 300 mg (0.3% hypericin); tea	Commonly 300 mg PO TID; not FDA-approved for treatment	Self-treated mild depression (limited evidence)
Ginkgo biloba	**Herbal supplement**	Capsule/tablet 40-120 mg	120-240 mg/day PO in divided doses; not FDA-approved for treatment	Self-treated memory complaints (limited evidence)
Fish oil (omega-3 fatty acids)	**Dietary supplement**	Capsule 300-1,000 mg EPA/DHA; Rx icosapent ethyl 1 g	1-4 g/day EPA plus DHA PO; Rx forms for hypertriglyceridemia	Hypertriglyceridemia, cardiovascular risk reduction (Rx)

How to tell them apart. Sources: S81; S82; S83

Herbal and dietary supplement profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
St. John's wort (Hypericum perforatum)	Photosensitivity, GI upset, dizziness, dry mouth	With SSRIs, transplant drugs, HIV protease inhibitors, oral contraceptives, warfarin	Serotonin syndrome with serotonergic drugs; lowers levels of warfarin, cyclosporine, contraceptives
Ginkgo biloba	Headache, GI upset, dizziness	Bleeding disorders; stop before surgery	Increased bleeding with warfarin, antiplatelets, NSAIDs; seizures with seed ingestion
Fish oil (omega-3 fatty acids)	Fishy aftertaste, belching, diarrhea	Hypersensitivity; fish or shellfish allergy (caution)	Bleeding risk at high doses with anticoagulants; atrial fibrillation at high doses

How to tell them apart. Sources: S81; S82; S83

Supporting drugs at a glance (listed for context; not profiled in detail)

Drug(s)	Why it matters
Vancomycin and fidaxomicin	Oral treatment for C. difficile infection
Vitamin K and 4-factor PCC	Reverse warfarin
Clindamycin	Antibiotic classically linked to C. difficile
Metformin and PPIs	Long-term use linked to low B12
Naloxone	Reverses opioid effects in loperamide or diphenoxylate overdose

Drug(s)	Why it matters
Cyclosporine, oral contraceptives, SSRIs	Interact with St. John's wort
Allopurinol and febuxostat	Block thiopurine metabolism; allopurinol needs a major dose cut, febuxostat is avoided
Methotrexate	Folate antagonist; folic acid is given alongside
Deferoxamine	Chelator used for iron overdose
Ketoconazole	CYP3A4 inhibitor that raises budesonide exposure

How to tell them apart. Sources: S84; S84 (notes)

Dietary supplements in the reading (textbook)

Supplement	Pharmacologic effects & clinical uses (clinical trials)	Drug interactions, precautions & dosag
Historic & regulatory factors	DSHEA (1994): regulated as foods; GMP rule only in 2007	FDA must prove harm to restrict; serious adverse events must be reported
Echinacea	Cold prevention (mixed trials)	Avoid in immune deficiency, autoimmune disease, immunosuppression; ≤ 10–14 days
Garlic	Modest lipid lowering	Antiplatelet: caution with warfarin, aspirin, ibuprofen
Ginkgo	Dementia (240 mg/d extract)	Antiplatelet: avoid with anticoagulants; avoid in seizure disorders
Ginseng	Glucose, fatigue	Antiplatelet; mania with phenelzine, lithium, neuroleptics
Milk thistle	Liver disease (no proven benefit)	No significant drug interactions
St. John's wort	Mild-moderate depression (900 mg/d)	CYP/P-gp induction; serotonin syndrome
Saw palmetto	BPH symptoms (160 mg twice daily)	No drug interactions reported
Purified nutritional supplements: coenzyme Q10	Low endogenous levels; heart failure	Vitamin K–like: may decrease warfarin effect
Glucosamine	Osteoarthritis (1500 mg daily)	May increase INR with warfarin
Melatonin	Jet lag (0.5–5 mg), insomnia	CYP1A2 (fluvoxamine); may decrease warfarin effect

How to tell them apart. Sources: p. 1204; p. 1207; p. 1208; p. 1210; p. 1211; p. 1213; p. 1214; p. 1215; p. 1216; p. 1217; p. 1218

WHY IT HAPPENS

- Why tell warfarin patients to keep vitamin K steady rather than avoid it? INR is dosed against their usual intake; swings, not the vitamin itself, cause trouble. S35

MNEMONIC (MEMORY AID)

- Fat-soluble **ADEK** stay; water-soluble **B and C** wash away. S30; SG U3-R12

HIGH-YIELD

- Nutrition and dietary supplements are not inert — vitamin status, drug-nutrient interactions, and herbal supplements can all meaningfully change drug therapy outcomes. S48; S40
- St. John's wort induces CYP3A4 → warfarin cleared faster → lower INR. S34; SG U3-R13
- Long-term PPI or metformin → check B12. S33; SG U3-R12
- Refeeding: start slowly, check phosphate/potassium/magnesium, thiamine first. S39

4 PART 4 · TOPIC

Putting It All Together: GI & Nutrition Toolkit

SO5 IO3.3 IO3.4

KEY DEFINITIONS

- Applying the whole GI & Nutrition toolkit: for each patient, name a first-line approach AND the single biggest safety concern to check before prescribing/counseling. S44; S44 (notes); S45 (notes)

HOW IT WORKS

- **1.** Acid-peptic disease and antiemetics are managed by targeting specific points along well-defined pathways (parietal cell secretion; the vomiting reflex). S48
- **2.** PPIs are the most potent antisecretory agents, but long-term use carries real risks that warrant periodic reassessment. S48
- **3.** IBD therapy follows a step-wise escalation matched to disease severity. **4.** Biologic agents share mechanisms and immunosuppression risk across organ systems. **5.** Nutrition and supplements are not inert. S48
- Pharmacology is cumulative, not siloed by unit: biologic mechanisms (Lecture 2) and drug-metabolism/interaction concepts (Lecture 1) reappear here in a new organ system. S48 (notes)

DRUGS NAMED AS EXAMPLES

- **Recall Q1:** PPIs block the H^+/K^+-ATPase proton pump itself — the final common pathway for gastrin, histamine, and acetylcholine — unlike H2RAs, which block only the histamine input. S50; S52
- **Recall Q2:** recent macrolide exposure raises clarithromycin-resistance concern, favoring bismuth quadruple therapy over standard clarithromycin-based triple therapy. S50; S52
- **Recall Q3:** aprepitant blocks NK1; ondansetron 5-HT3; meclizine H1/M1; metoclopramide D2. S50; S52

- **Recall Q4:** antimotility agents can worsen infectious/inflammatory diarrhea like C. difficile by delaying toxin clearance and slowing transit — toxic megacolon risk. S51; S52
- **Recall Q5–6:** vedolizumab blocks α4β7 gut-homing (most gut-selective); St. John's Wort induces CYP3A4, increasing warfarin metabolism and lowering INR — a pharmacokinetic drug-herb interaction. S51; S52

CLINICAL

- **Patient A — GERD symptoms, needs rapid occasional relief:** an antacid for occasional relief; an H2 blocker or PPI if frequent. S45; S45 (notes)
- **Patient B — confirmed H. pylori, penicillin allergy:** bismuth quadruple therapy. S45; S45 (notes)
- **Patient C — moderate-to-severe Crohn disease, inadequately controlled on azathioprine:** a biologic or small molecule such as an anti-TNF agent. S45; S45 (notes)
- **Patient D — warfarin patient starting an herbal mood supplement:** counsel on St. John's wort and check INR. S45; S45 (notes)

TRAP

- **Integrated case:** H. pylori + penicillin allergy + macrolide two months ago → bismuth quadruple therapy (allergy rules out amoxicillin triple therapy; macrolide exposure raises clarithromycin-resistance concern). SG U3-S
- Her subtherapeutic INR is St. John's Wort inducing CYP3A4 — stop the supplement (or adjust warfarin) with close INR monitoring. SG U3-S

Which drug would you choose? (S45)

Patient	First-line approach	Biggest safety check
A · GERD, rapid occasional relief	Antacid (H2 blocker or PPI if frequent)	Separate antacids from other drugs by 2 h; long-term PPI risks
B · H. pylori, penicillin allergy	Bismuth quadruple therapy	No amoxicillin; test of cure off PPI
C · Moderate-severe Crohn on azathioprine	Biologic or small molecule (anti-TNF)	TB and hepatitis screening; vaccines first
D · Warfarin + herbal mood supplement	Counsel on St. John's wort	Check INR

How to tell them apart. Sources: S45; S45 (notes)

MNEMONIC (MEMORY AID)

- Every case, two answers: **the approach** and **the one thing to check**. S45 (notes)

HIGH-YIELD

- Five takeaways: pathway targets · PPI risks · IBD step therapy · shared biologic risk · supplements matter. S48

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Laxative buckets	Bulk-forming · osmotic · stimulant · secretagogue S4
Fastest laxatives	Osmotic (~2 h), stimulant (~8 h); bulk-forming ~60 h S6; SG U3-R9
PEG	First-line chronic constipation S5
Lubiprostone vs. linaclotide	ClC-2, nausea · GC-C, diarrhea, not under age 2 S5; S60; S61
Loperamide	Peripheral mu agonist; P-gp keeps it out of CNS; high dose → QT S9; S10
Lomotil atropine	Subtherapeutic, deters misuse; Schedule V S9; SG U3-R10
C. difficile	No antimotility agents; oral vancomycin or fidaxomicin S11; SG U3-R10
IBD ladder	5-ASA → steroids (flare) → thiopurines (maintenance) → biologics S15; SG U3-R11
Sulfasalazine	Sulfapyridine AEs (rash, G6PD hemolysis, agranulocytosis); folic acid S16; S17
Mesalamine	Mild-moderate UC first-line; monitor renal function S16; S17
Budesonide	High first-pass → less systemic exposure S18
Thiopurines	TPMT/NUDT15 first; CBC, LFTs; allopurinol → cut dose 2/3–3/4 S19
IBD targets	TNF-α infliximab/adalimumab · α4β7 vedolizumab · IL-12/23 ustekinumab · JAK tofacitinib S20; S21
Gut-selective biologic	Vedolizumab S52; SG U3-R11
Before biologics	TB, hepatitis B/C, vaccines (live) first S24
Fat- vs. water-soluble	ADEK stored, toxic risk · B, C excreted S30; SG U3-R12
B12 risk	PPI/H2 blocker, metformin, bariatric surgery, older age, vegan S33
Iron absorption	↑ vitamin C, ↓ antacids/PPIs S32
Enteral vs. parenteral	If the gut works, use it S38
Refeeding	Slow start, thiamine, daily phosphate/K/Mg S39
Warfarin + vitamin K	Keep intake consistent S35
Warfarin + St. John's wort	CYP3A4 induction → ↓ INR S34; SG U3-R13

Prompt	Answer
Bleeding supplements	Ginkgo, fish oil S35; S40
Opioid-induced constipation	Methylnaltrexone, naloxegol, naldemedine (peripheral μ blockers) p. 1166
Sulfasalazine intolerance	Up to 40%; folic acid 1 mg/d p. 1174
TPMT	Low in 11%, absent in 0.3% — absent = no thiopurine p. 1175
Tofacitinib extras	Zoster ×4 (vaccinate), lipids at 8 weeks p. 1178
St. John's wort mechanism	PXR → CYP 3A4, 2C9, 1A2 + P-gp induction p. 1214
Antiplatelet herbs	Garlic, ginkgo, ginseng p. 1208; p. 1210; p. 1211
Warfarin and supplements	CoQ10, melatonin ↓ effect; glucosamine ↑ INR p. 1216; p. 1218

Self-test

First, the lecture's own 13 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S7-8

Based on Figure 2, a hospitalized patient needs constipation relief within a few hours. Which class is the most appropriate choice?

- **A** Bulk-forming
- **B** Stool softener
- **C** Osmotic or stimulant
- **D** None of these act quickly enough

ANSWER: C

Figure 2 shows osmotic agents (~2 hours) and stimulant agents (~8 hours) acting far faster than bulk-forming agents or stool softeners (~2–2.5 days) — appropriate choices when relief is needed within hours rather than days. S7-8

Lecture check 2 · S12-13

Why is a subtherapeutic dose of atropine included in the diphenoxylate/atropine combination product?

- **A** To treat the underlying cause of diarrhea
- **B** To discourage intentional overdose/misuse of the opioid component
- **C** To increase the antidiarrheal effect of diphenoxylate
- **D** To prevent constipation

ANSWER: B

At therapeutic doses, the atropine dose is too low to produce significant anticholinergic effects, but at supratherapeutic (misuse) doses, unpleasant anticholinergic effects (tachycardia, dry mouth, flushing) emerge — an intentional deterrent against abuse of the opioid agonist diphenoxylate. S12-13

Lecture check 3 · S22-23

In Figure 7, which biologic agent acts by a mechanism that is gut-selective rather than systemically immunosuppressive in the same way as the others?

- **A** Infliximab
- **B** Adalimumab
- **C** Vedolizumab
- **D** Tofacitinib

ANSWER: C

Vedolizumab blocks α4β7 integrin, which is relatively selective for gut-homing leukocyte trafficking — giving it a more gut-selective safety profile compared with TNF-α inhibitors or JAK inhibitors, which act more broadly on systemic immune signaling. S22-23

Lecture check 4 · S26-27

A patient with Crohn disease has been in remission on azathioprine for maintenance but now has an acute flare. What is the most appropriate immediate addition to therapy?

- **A** Increase the azathioprine dose only
- **B** Add a short course of corticosteroids to induce remission
- **C** Switch directly to a stimulant laxative
- **D** Discontinue all therapy until symptoms resolve on their own

ANSWER: B

Immunomodulators like azathioprine have a slow onset and are not effective for acute flares. Corticosteroids provide faster induction of remission during a flare, while the immunomodulator continues for long-term maintenance. S26-27

Lecture check 5 · S36-37

Based on Figure 10, a patient stable on warfarin starts taking a St. John's Wort supplement for mood. What is the expected effect on anticoagulation?

- **A** Decreased anticoagulant effect, due to increased warfarin metabolism via CYP3A4 induction
- **B** Increased anticoagulant effect, due to CYP3A4 inhibition
- **C** No effect, since herbal supplements do not interact with prescription drugs
- **D** Increased vitamin K levels directly counteracting warfarin

ANSWER: A

St. John's Wort is a well-known CYP3A4 inducer, which increases warfarin metabolism and lowers its anticoagulant effect (a pharmacokinetic interaction) — a distinct mechanism from dietary vitamin K, which opposes warfarin pharmacodynamically by supporting clotting factor synthesis. S36-37

Lecture check 6 · S42-43

A patient on long-term omeprazole therapy is found to have megaloblastic anemia and neurologic symptoms. Which deficiency is most likely, and why?

- **A** Vitamin B12 deficiency, due to reduced acid-dependent absorption
- **B** Vitamin C deficiency, due to increased renal excretion
- **C** Vitamin K deficiency, due to reduced dietary intake
- **D** Vitamin A deficiency, due to fat malabsorption

ANSWER: A

Long-term PPI use reduces gastric acid, which is needed to release B12 from dietary protein for absorption. This is one of the well-documented long-term PPI adverse effects covered in Session 1 and explains the megaloblastic anemia and neurologic symptoms here. S42-43

Lecture check 7 · S46-47

A patient with well-controlled GERD on a PPI for three years asks whether they should keep taking it indefinitely. What is the most appropriate response?

- **A** Periodically reassess the need for continued therapy, given long-term risks such as B12 deficiency, hypomagnesemia, fracture, and C. difficile risk
- **B** PPIs have no long-term risks, so continuing indefinitely is always fine
- **C** PPIs should never be used for more than one week
- **D** Switch immediately to an antacid with no further discussion

ANSWER: B

Long-term PPI use carries real, well-documented risks. Best practice is not indefinite, unexamined continuation — it is periodic reassessment of ongoing need, attempting dose reduction or discontinuation when clinically appropriate, while still treating confirmed indications adequately. S46-47

Lecture check 8 · S50; S52

Why do PPIs provide more complete acid suppression than H2 receptor antagonists?

- **A** They block only the gastrin pathway
- **B** They block the final common pathway (the proton pump) regardless of upstream signal
- **C** They neutralize acid directly, like antacids
- **D** They have no mechanistic advantage

ANSWER: B

PPIs block the H^+/K^+-ATPase proton pump itself — the final common pathway for gastrin, histamine, and acetylcholine signals — unlike H2RAs, which block only the histamine input. S50; S52

Lecture check 9 · S50; S52

A patient recently exposed to a macrolide antibiotic needs H. pylori eradication. Which regimen best avoids the concern this history raises?

- **A** Standard clarithromycin-based triple therapy
- **B** Bismuth quadruple therapy
- **C** Antacids alone
- **D** PPI monotherapy

ANSWER: B

Recent macrolide exposure raises concern for clarithromycin resistance, favoring bismuth quadruple therapy over standard clarithromycin-based triple therapy. S50; S52

Lecture check 10 · S50; S52

Which pairing correctly matches an antiemetic with its receptor target?

- **A** Ondansetron — D2 receptor
- **B** Aprepitant — NK1 receptor
- **C** Meclizine — 5-HT3 receptor
- **D** Metoclopramide — H1 receptor

ANSWER: B

Aprepitant blocks the NK1 receptor. Ondansetron blocks 5-HT3; meclizine blocks H1/M1; metoclopramide blocks D2. S50; S52

Lecture check 11 · S51; S52

A patient with suspected C. difficile infection asks for something to stop their diarrhea. What is the main concern with giving loperamide here?

- **A** Loperamide has no GI effects
- **B** Slowing transit can worsen toxin-mediated colonic injury
- **C** Loperamide treats the underlying infection
- **D** There is no concern in this scenario

ANSWER: B

Antimotility agents can worsen outcomes in infectious/inflammatory diarrhea like C. difficile by delaying clearance of toxin and slowing transit, raising risk of complications such as toxic megacolon. S51; S52

Lecture check 12 · S51; S52

Which IBD agent is most gut-selective in its mechanism, rather than broadly systemically immunosuppressive?

- **A** Infliximab
- **B** Vedolizumab
- **C** Tofacitinib
- **D** Azathioprine

ANSWER: B

Vedolizumab blocks α4β7 integrin-mediated leukocyte gut-homing, giving it a more gut-selective mechanism than TNF-α inhibitors, JAK inhibitors, or thiopurine immunomodulators. S51; S52

Lecture check 13 · S51; S52

A warfarin patient's INR drops after starting St. John's Wort. What mechanism explains this?

- **A** CYP3A4 induction, increasing warfarin metabolism
- **B** CYP3A4 inhibition, decreasing warfarin metabolism
- **C** Direct vitamin K supplementation
- **D** Increased GI absorption of warfarin

ANSWER: A

St. John's Wort is a CYP3A4 inducer, increasing warfarin metabolism and lowering its anticoagulant effect (and INR) — a pharmacokinetic drug-herb interaction. S51; S52

Original questions

Q1 · Mechanism · Laxatives & Antidiarrheals

Rank the laxative classes by onset, fastest first.

- **A** Bulk-forming, then osmotic, then stimulant
- **B** Secretagogue, then bulk-forming, then osmotic
- **C** All laxative classes act within the same hour
- **D** Osmotic/stimulant, then secretagogue, then bulk-forming
- **E** Stool softener, then bulk-forming, then stimulant

ANSWER: D

Osmotic and stimulant act within hours; bulk-forming and stool softeners take about 2–2.5 days (S6; SG U3-R9). S6; SG U3-R9

- **A.** Bulk-forming is slowest (S6).
- **B.** Osmotic is fastest (S6).
- **C.** Onset differs by mechanism (S6).
- **E.** Both of the first are slow (S6).

Q2 · Mechanism · Laxatives & Antidiarrheals

Which laxative works by activating guanylate cyclase-C to increase intestinal fluid secretion?

- **A** Senna
- **B** Linaclotide
- **C** Psyllium
- **D** Lubiprostone
- **E** Lactulose

ANSWER: B

Linaclotide is a guanylate cyclase-C agonist secretagogue (S61; S5). S5; S61

- **A.** Stimulant (S58).
- **C.** Bulk-forming (S55).
- **D.** Activates the ClC-2 chloride channel (S60).
- **E.** Osmotic (S57).

Q3 · Clinical vignette · Laxatives & Antidiarrheals

A patient needs relief from constipation tonight before a trip. Which class fits worst?

- **A** Stimulant (bisacodyl suppository)
- **B** Stimulant senna at bedtime
- **C** Bulk-forming psyllium
- **D** Osmotic polyethylene glycol
- **E** Osmotic magnesium hydroxide

ANSWER: C

Bulk-forming agents take the longest; a patient who needs relief tonight is not a bulk-forming candidate (S6 notes). S6

- **A.** Rectal bisacodyl works in 15–60 min (S59).
- **B.** 6–12 hours (S58).
- **D.** Fast, reliable (S4).
- **E.** Fast osmotic (S4).

Q4 · Clinical vignette · Inflammatory Bowel Disease

A Crohn patient on azathioprine develops gout and is started on allopurinol. What must happen?

- **A** Cut the azathioprine dose by two-thirds to three-quarters
- **B** Double the azathioprine to keep remission
- **C** Switch allopurinol to febuxostat freely
- **D** Stop azathioprine and start prednisone long term
- **E** No change — they use different enzymes

ANSWER: A

Allopurinol blocks thiopurine metabolism; reduce the thiopurine dose by about two-thirds to three-quarters (S19). S19; S84

- **B.** Exposure rises, not falls (S19).
- **C.** Febuxostat is avoided too (S84).
- **D.** Steroids aren't maintenance (S18).
- **E.** Xanthine oxidase clears thiopurines (S19).

Q5 · Mechanism · Inflammatory Bowel Disease

Why is budesonide associated with fewer systemic steroid effects than prednisone?

- **A** It is a 5-ASA prodrug cleaved in the colon
- **B** High first-pass metabolism limits systemic exposure
- **C** It is taken once monthly by injection
- **D** It has no glucocorticoid activity at all
- **E** It blocks α4β7 integrin in the gut

ANSWER: B

Budesonide has high first-pass metabolism — less systemic exposure than prednisone (S18). S18; S66

- **A.** That is sulfasalazine (S17).
- **C.** Oral daily (S66).
- **D.** It is a corticosteroid (S66).
- **E.** That is vedolizumab (S20).

Q6 · Clinical vignette · Inflammatory Bowel Disease

Before starting infliximab for Crohn disease, which steps are required?

- **A** TB, hepatitis B/C screening; vaccines first
- **B** TPMT and NUDT15 genotyping only, nothing else
- **C** G6PD activity testing before the infusion
- **D** A baseline DEXA scan of the spine only
- **E** Nothing until an infection actually appears

ANSWER: A

Screen for tuberculosis and hepatitis B/C, and update vaccines (especially live) before starting biologics (S24). S24

- **B.** That is for thiopurines (S19).
- **C.** Relates to sulfasalazine hemolysis (S17).
- **D.** Not the taught biologic step (S24).
- **E.** Screening comes first (S24).

Q7 · Mechanism · Nutrition, Vitamins & Dietary Supplements

Why are fat-soluble vitamins more prone to toxicity than water-soluble vitamins?

- **A** They are given only intravenously
- **B** They are stored in fat and liver and accumulate
- **C** They are absorbed without fat or bile
- **D** Deficiency develops faster without intake
- **E** They are excreted renally and build up in CKD

ANSWER: B

Fat-soluble vitamins (A, D, E, K) are stored and accumulate; water-soluble vitamins are renally excreted and rarely toxic (S30; SG U3-R12). S30; SG U3-R12

- **A.** Not the reason (S30).
- **C.** Absorption depends on fat and bile (S30).
- **D.** That describes water-soluble vitamins (S30).
- **E.** Water-soluble vitamins are excreted renally (S30).

Q8 · Clinical vignette · Nutrition, Vitamins & Dietary Supplements

A severely malnourished patient is about to start feeding. Which plan prevents refeeding syndrome?

- **A** Start slowly, give thiamine, check phosphate/K/Mg daily
- **B** Use parenteral nutrition even though the gut works
- **C** Check electrolytes once at discharge
- **D** Withhold thiamine until after feeding
- **E** Start at full calories to correct deficits fast

ANSWER: A

Start feeding slowly, give thiamine before feeding, and check phosphate, potassium and magnesium daily at first (S39). S39

- **B.** If the gut works, use it (S38).
- **C.** Daily checks at the start (S39).
- **D.** Thiamine goes before feeding (S39).
- **E.** Rapid carbohydrate load triggers the shifts (S39).

Q9 · Mechanism · Nutrition, Vitamins & Dietary Supplements

How do long-term PPIs cause B12 deficiency?

- **A** They bind B12 in the gut like antacids
- **B** They block ileal uptake like metformin
- **C** They destroy intrinsic factor in the ileum
- **D** They speed renal excretion of B12
- **E** Less acid frees less B12 from food protein

ANSWER: E

Stomach acid and pepsin release B12 from food protein; PPIs reduce acid (S33; SG U3-R12). S33; SG U3-R12

- **A.** Not the taught mechanism (S33).
- **B.** Metformin impairs ileal uptake (S33).
- **C.** Intrinsic factor comes from parietal cells; acid release is the issue (S33).
- **D.** Not the mechanism (S33).

Q10 · Clinical vignette · Nutrition, Vitamins & Dietary Supplements

A patient on warfarin asks whether to stop eating leafy greens. What is the right counseling?

- **A** Take ginkgo to balance the greens
- **B** Double greens on days INR is low
- **C** Diet never affects warfarin at all
- **D** Keep vitamin K intake consistent
- **E** Avoid all vitamin K foods completely

ANSWER: D

Keep vitamin K intake consistent rather than avoiding it; check INR when starting or stopping supplements (S35). S34; S35

- **A.** Ginkgo raises bleeding risk (S35).
- **B.** Swings destabilize INR (S35).
- **C.** Vitamin K opposes warfarin (S34).
- **E.** Consistency, not avoidance (S35).

Q11 · Clinical vignette · Nutrition, Vitamins & Dietary Supplements

A patient on warfarin starts glucosamine for knee arthritis. What does the textbook warn?

- **A** The INR will fall from CYP3A4 induction
- **B** No interaction — glucosamine is a food
- **C** The INR may rise, increasing bleeding risk
- **D** Serotonin syndrome with warfarin
- **E** Glucosamine supplies vitamin K and lowers INR

ANSWER: C

Glucosamine sulfate may increase the INR in patients taking warfarin; avoid or monitor carefully (p. 1216). p. 1216

- **A.** That is St. John’s wort (p. 1214).
- **B.** An INR interaction is reported (p. 1216).
- **D.** Not a glucosamine effect (p. 1216).
- **E.** CoQ10 is the vitamin K–like one (p. 1216).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO3.1** Identify major pharmacologic classes used in GI and nutrition conditions and describe their mechanisms of action.
- [ ] **IO3.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions of GI and nutrition medications.
- [ ] **IO3.3** Compare agents within and across these classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO3.4** Select appropriate pharmacologic therapy for common GI and nutrition conditions based on clinical presentation and patient-specific factors.
- [ ] **IO3.5** Identify appropriate dosing, administration, and monitoring considerations for GI and nutrition medications.
- [ ] **IO3.6** Explain important medication-use and safety information to patients receiving GI and nutrition pharmacotherapy.
- [ ] **SO1** Classify laxatives and antidiarrheal agents by mechanism and clinical use, including representative generic agents.
- [ ] **SO2** Explain the step-wise pharmacologic management of inflammatory bowel disease, including representative generic agents at each step.
- [ ] **SO3** Identify major pharmacologic classes used in nutrition support and describe clinically significant drug-nutrient and herbal-supplement interactions.
- [ ] **SO4** Identify key contraindications, adverse effects, and drug interactions throughout.
- [ ] **SO5** Apply the full GI & Nutrition pharmacology toolkit to integrated clinical scenarios.